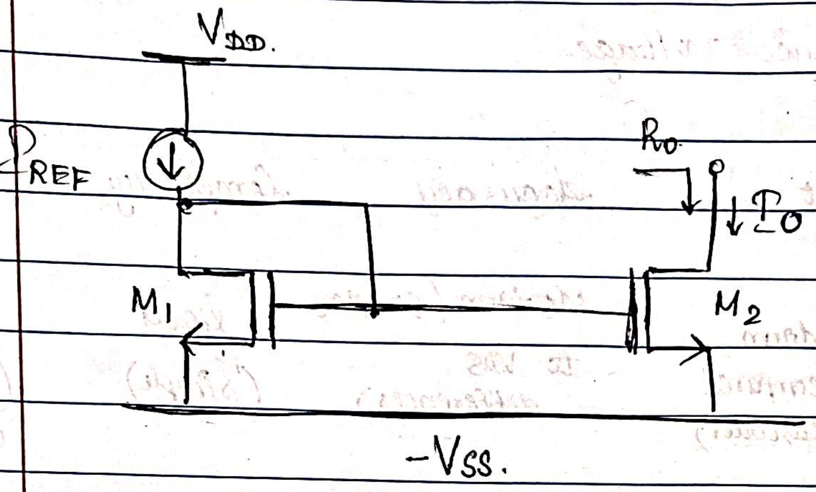
1) Make a table which lists the three current mirror topologies described in this lab. Rate each topology using good, medium and bad for the following design considerations: Rout, accuracy, complexity and compliance voltage.

Topology	Rout	Accuracy	Complexity	Compliance voltage.
1) Simple current mirror.	Bad → Medium (low compared with cascodes)	Medium (sensitive to VDS differences)	Good (simple)	Good (lowest compliance)
2) Cascode current mirror	Good (much higher r_out)	Good (better VDS matching)	Bad (higher complexity more transistors)	Bad (large compliance voltage)
3) Low-voltage Cascode current mirror.	Good → Medium (close to cascode for r_out)	Good	Medium (more complex than simple, less than full cascode	Medium (reduced compliance vs full cascode, typically 1 $V_{DS(sat)}$ lower than full cascode

2) Design a simple 1:1 current mirror that has a compliance voltage of 125 mV to 175 mV. The output current should be 120 μA. Target an output impedence of 25 kΩ or more. Determine W and L for each transistor



$I_O = 120\,\mu A$

$V_C = V_{DS} = 130\,mV$

From Lab 1, choose : $W = 0.5\,\mu$

1 fingers. $L =$ ~~0.18~~ $0.27\,\mu$

$\frac{I_O}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$

$V_{tn} = 447.0368\,mV$

$\mu_n C_{ox} = 214.326\,\mu A/V^2$

$\lambda = 0.20997$

$f_t = 34.1123$

$V_{GS} = 698.48\,mV$

$I_D = 120\,\mu A$

$I_0 = I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_2 (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$

$120\,\mu A = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_2 (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$

Choosing the values.

$r_{dsn} = \frac{1}{\lambda I_D} > 25\,k\Omega$

$\lambda \leq \frac{1}{25\,k\Omega \times 120 \times 10^{-6}}$

$\lambda \leq \frac{1}{\cancel{0.24}\,3}$

0.33

$\lambda \leq \cancel{4.16667}$

$\lambda \leq 0.333$

$$120\,\mu A = \frac{1}{2}\left(214.326\,\mu A/V^2\right) \times \left(\frac{W}{L}\right)_2 \times \left(698.48 - 447.0368\right)^2 mV \times \left(1 + (0.20997)\,130\,mV\right)$$

$$120 = \frac{1}{2}\left(214.326\right) \times \left(\frac{W}{L}\right)_2 \times \left(0.2514\right)^2 \times \left(1 + (0.20997)(0.13)\right)$$

$$120 = (26.9407) \times (1.027296) \left(\frac{W}{L}\right)_2 \times 0.2514$$

$$\left(\frac{W}{L}\right)_2 = \cancel{4.33587}\ 17.2469.$$

$L = 0.27\ \mu m.$

$W = \cancel{4.33587 \times 0.27\,\mu m}\ 17.2469 \times 0.27\,\mu m$

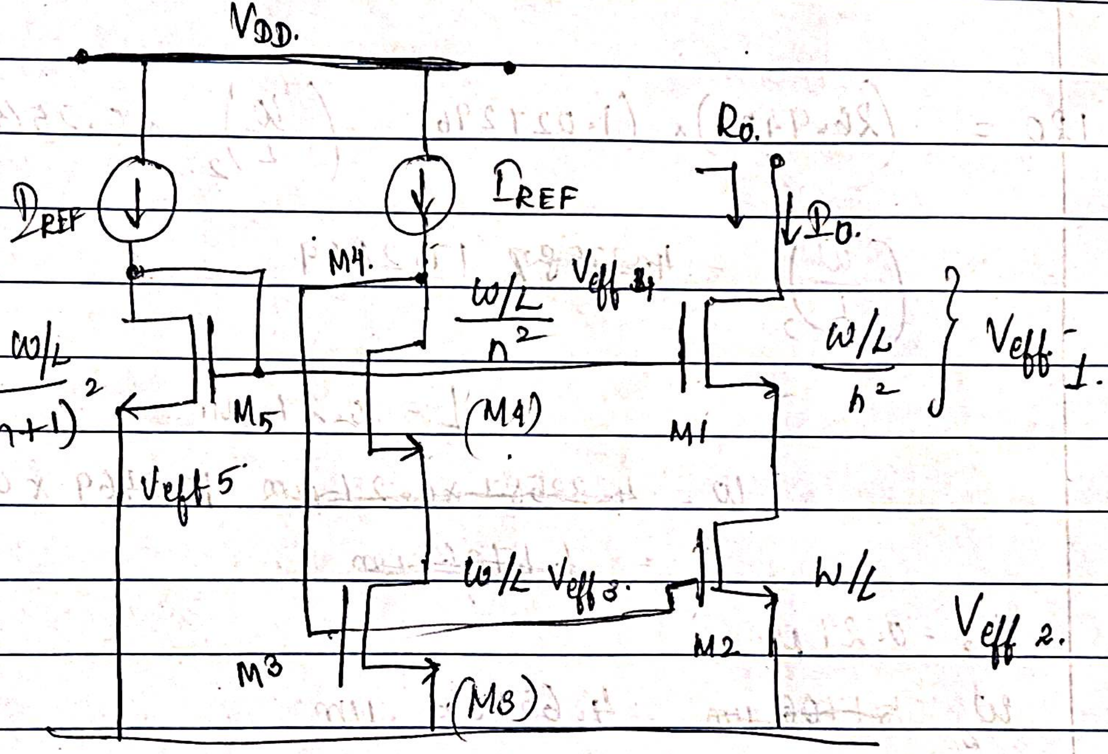
$= \cancel{1.1706\,\mu m}$

$L_2 = 0.27\ \mu m.$

$W_2 = \cancel{1.1706\,\mu m}\ \ 4.656663\ \mu m$

$$\boxed{\begin{aligned} L_2 &= 0.27\ \mu m. \\ W_2 &= 4.656663\ \mu m. \end{aligned}}$$

3) Design a low-voltage cascode current mirror with a 1:2 input to output current ratio. The compliance voltage should be less than 50mV. Determine W and L for each transistor and the value of the bias current. In your design calculations, you should provide validation that all devices will be in the saturation region for the circuit's DC operating point



Choose $n = 4$.

Let Compliance voltage $= 0.05\,V$
$= 50\,mV$.

$$V_{o\,(minimum)} = (n+1)\,V_{eff}$$

$$0.05\,V = (4+1)\,V_{eff}$$

$$\frac{0.05}{5} = V_{eff}$$

$$\Rightarrow V_{eff} = 0.01$$

$V_{eff1} = V_{eff4} = nV_{eff}$

$V_{eff1} = V_{eff4} = 0.01 \times 4. = 0.04.$

$V_{eff2} = V_{eff3} = V_{eff} = 0.01$

$V_{eff5} = (n+1)\ V_{eff} = (4+1)\ 0.01$

$= (5) \times 0.01$

$V_{eff5} = 0.05.$

$V_{eff1} = 0.04\ V$

$V_{eff2} = 0.01\ V.$

$V_{eff3} = 0.01\ V$

$V_{eff4} = 0.04\ V$

$V_{eff5} = 0.05\ V$

$V_{eff} = V_{GS} - V_t.$

$V_{GS} = V_{eff} + V_t.$

Now, finding the V_{GS} values.

Let, $V_t = 447.0368\ mV.$

$V_{GS1} = V_{eff1} + V_t.$

$= 0.04 + 0.4470368$

$= 0.487\ V$

$V_{GS2} = V_{eff2} + V_t$

$= 0.01 + 0.4470$

$= 0.457.\ V$

$V_{GS3} = V_{eff3} + V_t$

$= 0.01 + 0.4470368$

$= 0.4570$

$V_{GS4} = V_{eff4} + V_t$

$= 0.04 + 0.447$

$= 0.487\ V$

$V_{GS5} = V_{eff5} + V_t = 0.05 + 0.4470368$

$= 0.4970$

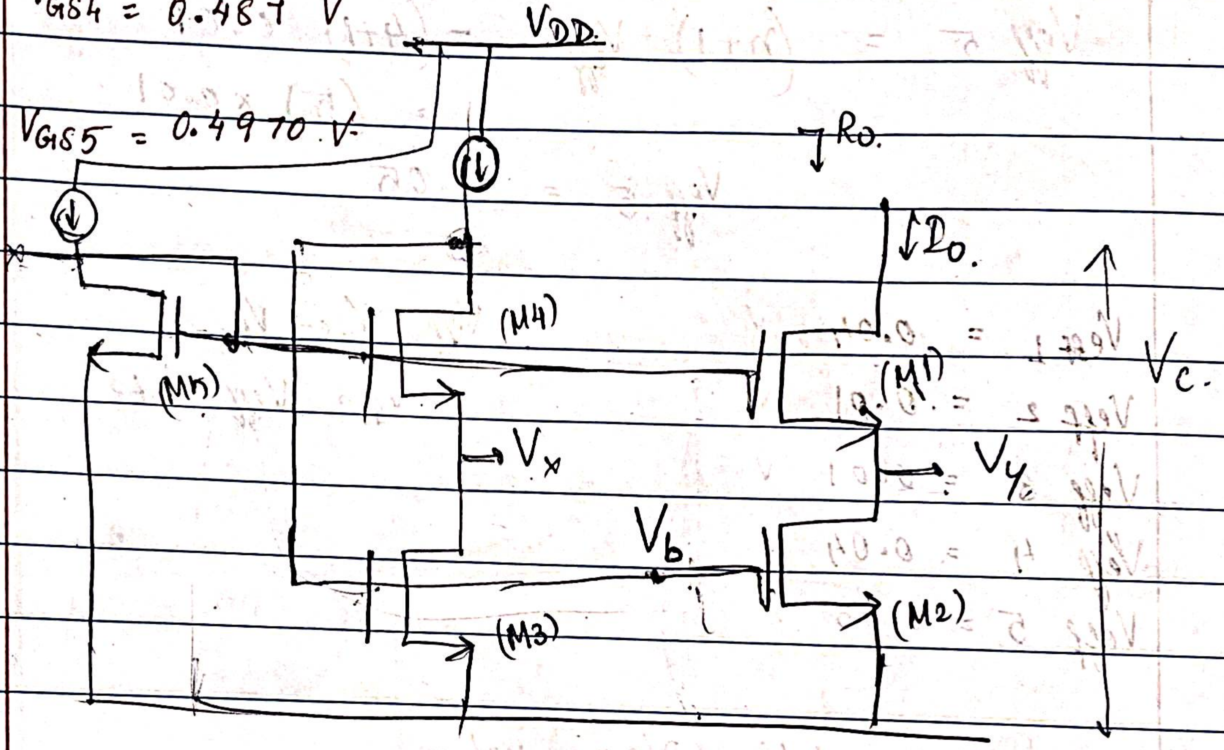
$V_{GS1} = 0.487\ V$

$V_{GS2} = 0.457\ V$

$V_{GS3} = 0.457\ V$

$V_{GS4} = 0.487\ V$

$V_{GS5} = 0.4970\ V$



$V_b = V_{GS2}$

$V_b = 0.457\ V$

$V_x = V_y = V_{eff}$

$V_{DS4} = V_b - V_{eff}$

$V_{DS4} = \cancel{V_{eff}}\ 0.457 - 0.01$

$\boxed{V_{DS4} = 0.447}$

$V_{DS3} = V_x = V_{eff}$

$\boxed{V_{DS3} = 0.01}$

$V_{DS2} = V_{effect} = V_y$

$\boxed{V_{DS2} = 0.01}$

$V_{DS1} = V_c - V_y$

$= 0.05 - 0.01$

$\boxed{V_{DS1} = 0.04}$

$V_{DS5} = V_{G5} = V_{G1}$

$= \cancel{0.4970}\ V$

$V_{GS1} = V_{G1} - V_t$

$V_{GS1} + V_{eff} = V_{G1}$

$\cancel{0.487 + V_{eff} = V_{G1}}$

$0.497 = V_{G1}$

$V_{DS1} = 0.04$

$V_{DS2} = 0.01$

$V_{DS3} = 0.01$

$V_{DS4} = 0.447$

$V_{DS5} = 0.4970$

Checking if transistors are in saturation region.

for M_1 = $V_{DS1} = 0.04$.

$V_{GS1} - V_{tn1} = 0.487 - 0.447$.

$= 0.04$. $\quad 0.04 > 0.04$.

for M_2 = $V_{DS2} = 0.01$

$V_{GS2} - V_{tn2} = 0.457 - 0.447 = 0.01$.

$0.01 > 0.01$

for $M_3 \Rightarrow V_{DS3} = 0.01$

$V_{GS3} - V_{tn3} = 0.457 - 0.4470 \quad 0.01 > 0.01$

$= 0.01$

~~for M_4 = $V_{DS4} = 0.441$~~

$V_{GS4} - V_t = 0.487 - 0.44703$

$= 0.03997$.

$0.447 > 0.0399$

~~for M_5 = $V_{DS5} = 0.4970$~~

$V_{GS5} - V_t$

$= 0.4970 - 0.4470$

$= 0.05$

$\boxed{0.497 > 0.05}$

Now, finding $\frac{W}{L}$ ratios.

$\mu_n C_{ox} = 214.326$

$\lambda = 0.20997$

$L = 0.27\mu$

for M1,

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_1 (V_{GS_1} - V_t)^2 (1 + \lambda V_{DS})$$

$$120\mu A = \frac{1}{2}(214.326)\left(\frac{W}{L}\right)_1 \times (0.487 - 0.447)^2 (1 + (0.20997 \times 0.04))$$

$$120\mu A = \cancel{107.17}\ 0.1729 \left(\frac{W}{L}\right)_1$$

$$694 = \left(\frac{W}{L}\right)_1 \Rightarrow \boxed{L = 0.27\mu}$$

$$\boxed{W = 187.88\mu.}$$

for M2,

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_1 (V_{GS_2} - V_t)^2 (1 + \lambda V_{DS2})$$

$$120^{\mu A} = \frac{1}{2}(214.326)\left(\frac{W}{L}\right)_2 (0.457 - 0.447)^2 (1 + (0.20997 \times 0.01))$$

$$120 = \cancel{0.010738} \left(\frac{W}{L}\right)_2 \times \cancel{0.021477}\ 0.010738.$$

$$\cancel{5587.21} = \left(\frac{W}{L}\right)_2 = 11175.265$$

$$\boxed{L = 0.27\mu.}$$

$$\boxed{W = \cancel{1508.54}\mu.}\ 3017.32\mu$$

for M3,

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS_3}-V_t)^2(1+\lambda V_{DS3})$$

$$= \frac{1}{2}(214.326)\left(\frac{W}{L}\right)(0.457-0.441)^2(1+(0.20997 \times 0.01))$$

$$\cancel{120}\ 60 = 0.010738\left(\frac{W}{L}\right)$$

$$5587.632\ \cancel{11174.43184} = \left(\frac{W}{L}\right)$$

$$1508.6\mu = W \qquad L = 0.27\mu.$$

$$\boxed{\cancel{3017.09\mu m = W}}\ \boxed{W = 1508.6\mu.}$$

for M4,

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS_4}-V_t)^2(1+\lambda V_{DS4})$$

$$= \frac{1}{2}\times(214.326)\times\frac{W}{L}(0.487-0.441)^2(1+(0.20997\times 0.447))$$

$$60\ \cancel{120} = 0.18755\left(\frac{W}{L}\right)$$

$$319.914\ \cancel{639.829} = \left(\frac{W}{L}\right)$$

$$L = 0.27\mu.$$

$$W = 639.829 \times 0.27\mu$$

$$\boxed{\cancel{W = 172.753\mu}}$$

$$\boxed{W = 86.3769}$$

$\underline{M_5}$

$$I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{GS_5} - V_t)^2 (1 + \lambda V_{DS5})$$

$$60 = (0.5)(214.326) \times \left(\frac{W}{L}\right)_5 \left((0.4970 - 0.447)^2\right)\left(1 + (0.20997 \times 0.27)\right)$$

$$60 = \left(\frac{W}{L}\right)_5 \times 0.28309$$

$$211.942 = \left(\frac{W}{L}\right)_5 \quad L = 0.27\mu$$

$$\boxed{W_5 = 57.224\mu}$$

$W_1 = 187.38\mu$

$W_2 = \cancel{1508.54\mu}$ 8017.82μ

$W_3 = 1508.6\mu$

$W_4 = 86.8769\mu$

$W_5 = 57.224\mu$